Lunar Survival Activity

For the instructor

Goal Students decide which supplies matter most when lives depend on it, first alone and then as a group.

Time 20 to 30 minutes. No background knowledge needed.

Before class Print page 2, one copy per student, or one per pair or small group.

STEPS

1. Read the scenario aloud.
2. Give students 5 to 10 minutes to rank the items.
3. Read the NASA rankings from the answer sheet. Students write them in the NASA Ranking column.
4. To score, find how far each of the student's ranks is from NASA's rank, ignoring plus or minus. Add up the differences. The lower the total, the better.

SCORE BANDS

0-25	Excellent
26-32	Good
33-45	Average
46-55	Fair
56-70	Poor
71-112	Very Poor

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Your space crew was supposed to land beside the mothership on the moon. A mechanical failure forced your ship down 200 miles away, on the sunlit side of the moon, and the landing damaged most of the equipment on board. The 15 items below came through undamaged. Rank them by how important they are for your crew's 200-mile trip back to the mothership. Number 1 is the most important and 15 the least. The list is in no particular order.

Your Ranking	Item	NASA Ranking
	Box of matches	
	Food concentrate	
	50 feet of nylon rope	
	Parachute silk	
	Portable heating unit	
	Two .45 Caliber pistols	
	One case of dehydrated milk	
	Two 100 lb. tanks of oxygen	
	Stellar Map	
	Self-inflating life raft	
	Magnetic compass	
	20 liters of water	
	Signal flares	
	First aid kit, including injection needle	
	Solar powered FM receiver-transmitter	

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Answer Sheet

NASA Rank	Item	Reason
1	**Two 100 lb. tanks of oxygen**	There is no air to breathe on the moon.
2	**20 liters of water**	Needed to survive, like oxygen.
3	**Stellar Map**	The main way to find your direction.
4	**Food concentrate**	A compact way to get energy.
5	**Solar powered FM receiver-transmitter**	To talk with the mothership.
6	**50 feet of nylon rope**	For climbing and for tying injured crew members together.
7	**First aid kit, including injection needle**	To treat injuries.
8	**Parachute silk**	Shade from the sun.
9	**Self-inflating life raft**	Its CO_2 bottle might push you along.
10	**Signal flares**	To signal the mothership once it is in sight.
11	**Two .45 Caliber pistols**	Possible self-propulsion.
12	**One case of dehydrated milk**	Food, much like the concentrate.
13	**Portable heating unit**	Not needed on the sunlit side of the moon.
14	**Magnetic compass**	Useless: the moon has no magnetic poles to point to.
15	**Box of matches**	Useless: the moon has no oxygen for a flame.

Rankings from NASA's classic "Survival on the Moon" exercise. Use it to practice reasoning as a team, not as real survival advice.

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